Andrea Moschetto

Data Science Student and Software Engineer
Catania, Sicily, Italy

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Summary

Recent M.Sc. graduate in Computer Science with a strong foundation in Artificial Intelligence, Computer Vision, and Software Engineering. I specialize in bridging the gap between advanced theoretical research—highlighted by 3 publications in ML and Algorithms—and scalable, production-ready applications. Leveraging a deep passion for coding and hands-on experience in backend development and Python microservices, I am focused on building, optimizing, and deploying robust Machine Learning solutions in real-world environments.

Publications

2025 Andrea Moschetto, Lemuel Puglisi, Alec Sargood, Pierluigi Dell'Acqua, Francesco Guarnera, Sebastiano Battiato and Daniele Ravì. Benchmarking GANs, Diffusion Models, and Flow Matching for T1w-to-T2w MRI Translation. CVGMMI Workshop (ICIAP 2025).

2024 Simone Faro, Francesco Pio Marino, and Andrea Moschetto. Beyond Horspool: A Comparative Analysis in Sampled Matching. In *Prague Stringology Conference 2024*, 16 (2024).

2024 Simone Faro, Francesco Pio Marino, Andrea Moschetto, Arianna Pavone, and Antonio Scardace. The Great Textual Hoax: Boosting Sampled String Matching with Fake Samples. In 12th International Conference on Fun with Algorithms (FUN 2024). LIPIcs, Volume 291, pp. 13:1-13:17, Schloss Dagstuhl – Leibniz-Zentrum für Informatik (2024)

Work Experience

Computer Science Tutor — Oct 2024 – Jul 2025
University of Catania — *Catania, Sicily, Italy*

- Tutoring activities for the course *Fondamenti di Informatica*
- Topics: Computational Models, Formal Languages, Functional Programming, Logic

Software Engineer and Developer — Jan 2023 – Aug 2023
Herakel — *Remote – Catania, Sicily, Italy*

- Developed distributed systems and microservices using Python
- Designed event-driven architectures with Kafka
- Implemented REST APIs using FastAPI and integrated with external services
- Wrote integration and unit tests to ensure software reliability

Education

M.Sc. in Computer Science, University of Catania — Oct 2024 – Apr 2026
Catania, Sicily, Italy
Grade: 110/110 cum Laude + nomination for the Archimedes Award
Thesis: Inverting Task Graph Learning: Differentiable Sequence Optimization for Video Understanding and Topological Sorting

B.Sc. in Computer Science, University of Catania — Oct 2018 – Mar 2024
Catania, Sicily, Italy
Grade: 110/110 cum Laude
Thesis: Efficient Online String Matching Based on Character Distance Sampling

Archimede Technical Institute Sep 2013 – Jul 2018
Catania, Sicily, Italy *Diploma in Technical Computer Science*

Awards

- 2018/19 Academic Scholarship, ERSU
- 2019/20 Academic Scholarship, ERSU
- 2020/21 Academic Scholarship, ERSU

Interests and Studies

- Machine Learning, Deep Learning, AI Safety, Explainability, Model Evaluation
- Large Scale Software Engineering, Distributed Systems, Design Patterns
- Computer Vision, Video Understanding, Self-Supervised Learning
- Algorithms, String Matching, Optimization, Computational Complexity

Technical Skills

Languages	:	C, C++, C#, Java, JavaScript, Typescript, Python, PHP, Erlang, Prolog
AI / ML	:	PyTorch, Lightning, Pandas, NumPy, Scikit-learn, Matplotlib, OpenCV
WebDev	:	FastAPI, Alembic, PyTest, Laravel, Express, Node.js, HTML, CSS, SQL, Apache
Graph-Engines	:	Unity
Dev Ops	:	AWS, Docker, Docker-Compose, Kafka, Kubernetes, GitHub Actions
Dev Tools	:	VS Code, Git, GitLab, GitHub

I authorize the processing of my personal data according to EU Regulation 2016/679 (GDPR) and Italian Legislative Decree 101/2018 for the purposes of personnel recruitment and selection.

Andrea Moschetto Apr 2026